



Feature Extraction

ESMC Protein Encoder

960 dim

SMILES_Transform Encoder

Tokenization + Embedding

1024 dim

Concat Joint Feature (1,984 dim)

Predictor

Shared Projection (MLP)

Linear - BN - ReLU - Dropout (hidden_dim, configurable)

Token Projection (attn_tokens x attn_dim)

Multi-Head Attention (attn_heads)

Gated Residual Addition (attn_dim)

Residual Bypass

Linear (Km Head, output=1)

Linear (kcat Head, output=1)